A Tiny Robot That Swims With Living Muscles

From: ScienceDaily All Mined: 2026-10-23 5 min read

Most swimming robots carry a motor. A new robot from MIT moves through the water in a different way: it is powered by a thin layer of living muscle cells.

The tiny robot has a soft gel body about the size of a stick of gum, with two fins coated in muscle cells far thinner than a human hair. The cells were **genetically** changed so they squeeze, or **contract**, whenever light shines on them.

By aiming light at one fin or the other, the researchers can make the fins flap and push the robot forward. Shining light more or less often, and choosing which fin gets it, lets the team steer the robot and control its speed.

In experiments, the robot swam and turned through an underwater maze. At its fastest, it covered about four times its own body length in one minute. That is much slower than an Olympic swimmer, who can travel up to 65 body lengths per minute, but about the pace of a cow shark cruising through the ocean.

'It takes a lot of force to move through water versus air,' said study author Ritu Raman, an associate professor of mechanical engineering at MIT. 'The robot's quite strong, given its size.'

The study, published in the journal Advanced Functional Materials, is the first demonstration of a very thin robot that can travel from place to place using living muscle **tissue**. Machines that mix living **tissue** with human-made materials are called **biohybrid** robots.

Raman says thin designs could be cheaper to build and move more efficiently than bulky **biohybrid** robots made from millions of cells. One day, robots powered by living muscle might do **delicate** jobs in places too fragile for ordinary machines, because living **tissue** is soft, responds to its surroundings, and can heal itself.

Word Treasure

contract -- To squeeze or shorten, as the muscle cells do when light shines on them.

biohybrid -- Made by mixing living tissue with human-made materials.

genetically -- Relating to genes, the instructions inside cells that scientists changed in this study.

tissue -- A group of similar living cells that together form part of a body.

delicate -- Easily damaged and needing gentle, careful handling.

Background you need

Biohybrid robots mix living tissue with human-made materials, so a machine can move using real muscle instead of only a motor.

Muscle cells move by contracting, and scientists can control specially changed cells with light, turning light signals into movement.

Soft, living materials can be gentler and can heal themselves, which is why researchers explore them for delicate jobs in fragile places.

Break it down

LEAD: A tiny MIT robot swims using living muscle cells instead of a motor

KEY EVENT: Researchers steered the gum-sized robot through an underwater maze using light

+- WHO: MIT researchers, including study author Ritu Raman

+- HOW: Light makes the muscle cells on a fin contract and flap

EVIDENCE: At its fastest the robot covered about four times its body length in one minute

OPEN QUESTION: Whether thin biohybrid robots will one day do delicate jobs in fragile places

Why it matters

Mixing living tissue with machines could create softer robots that move efficiently and work safely in fragile places that ordinary machines might damage.

Test what you remember

1. What powers the MIT swimming robot instead of a motor?

- (A) A battery and a propeller
- (B) Magnets in the water
- (C) A thin layer of living muscle cells
- (D) A tiny sail

2. What do the muscle cells do when light shines on them?

- (A) They glow green
- (B) They squeeze, or contract
- (C) They melt
- (D) They grow larger

3. How do the researchers steer the robot?

- (A) By aiming light at one fin or the other
- (B) With spoken commands
- (C) With a remote-control wheel
- (D) By changing the water temperature

4. At its fastest, how far did the robot travel in one minute?

- (A) About 65 times its body length
- (B) About ten body lengths
- (C) About one body length
- (D) About four times its own body length

5. What are machines that mix living tissue with human-made materials called?

- (A) Gel robots
- (B) Light bots
- (C) Muscle machines
- (D) Biohybrid robots

6. Why might living-muscle robots be useful one day?

- (A) Living tissue is soft, responds to its surroundings and can heal itself
- (B) They are made of metal
- (C) They can swim faster than any fish
- (D) They never need light

Answer key (try the quiz first!): 1: C 2: B 3: A 4: D 5: D 6: A

Questions to think about

1. Ritu Raman: She says it takes a lot of force to move through water, the robot is quite strong for its size, and thin designs could be cheaper to build and move more efficiently than bulky biohybrid robots.

MY THOUGHTS (at least 20 words):